Which lymphoid cancers should learn from each other?

Request for review of a list fixed before testing

Draft for the study haematologists. Fill in this PDF directly (Adobe Reader or Preview), save it and send it to Mikkel Werling.

Your name:

Background. We have built one prediction model for patients starting first-line treatment for any of 24 lymphoid cancers (37,531 Danish patients). For each patient it predicts the risk of 69 outcomes over the following two years, from relapse and second-line treatment to infection, cytopenia, organ toxicity and hospital admission. Because the model learns from all diseases at once, what it learns in one disease can be used for another.

What we will test. After the model is trained, we can switch off what it learned from one disease and see how much worse its predictions become for another. We want to compare this with clinical knowledge, and that comparison is only fair if the clinical expectations are written down **before** we look at the results on the test patients. Your ratings are those expectations.

What we ask of you.

1. For each pair below: how much would you expect what is learned in the first disease to help predict the stated outcomes in the second? Choose *strong*, *moderate*, *weak*, *none* or *don't know*, and add a comment where useful. The pairs are in random order, and some may well deserve *none*.
2. Suggest pairs we have missed. Any disease can be paired with any other, for any group of outcomes; the diseases and outcomes are listed at the end.

	Pair	For predicting	Expected help	Comment
P1	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Small lymphocytic lymphoma (SLL)**	all outcomes	strong moderate weak none don't know	
P2	Learn from **Peripheral T-cell lymphoma, NOS (PTCL)** to predict for **T-cell lymphoma, unspecified (TCL)**	disease course and later malignancy	strong moderate weak none don't know	
P3	Learn from **Multiple myeloma (MM)** to predict for **Chronic lymphocytic leukaemia (CLL)**	infection and antimicrobial use	strong moderate weak none don't know	
P4	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Multiple myeloma (MM)**	infection and antimicrobial use	strong moderate weak none don't know	
P5	Learn from **Hairy-cell leukaemia (HCL)** to predict for **Diffuse large B-cell lymphoma (DLBCL)**	disease course and later malignancy	strong moderate weak none don't know	

	Pair	For predicting	Expected help	Comment
P6	Learn from **Burkitt lymphoma (BL)** to predict for **Lymphoblastic lymphoma (LBL)**	neutrophil, white-cell and platelet findings	strong moderate weak none don't know	
P7	Learn from **Peripheral T-cell lymphoma, NOS (PTCL)** to predict for **Angioimmunoblastic T-cell lymphoma (AITL)**	disease course and later malignancy	strong moderate weak none don't know	
P8	Learn from **Anaplastic large-cell lymphoma (ALCL)** to predict for **Chronic lymphocytic leukaemia (CLL)**	all outcomes	strong moderate weak none don't know	
P9	Learn from **Multiple myeloma (MM)** to predict for **Hodgkin lymphoma (HL)**	all outcomes	strong moderate weak none don't know	
P10	Learn from **Chronic lymphocytic leukaemia (CLL)** to predict for **Lymphoplasmacytic lymphoma (LPL)**	cardiovascular and thrombotic events	strong moderate weak none don't know	
P11	Learn from **Burkitt lymphoma (BL)** to predict for **Lymphoblastic lymphoma (LBL)**	metabolic and cell-turnover markers	strong moderate weak none don't know	
P12	Learn from **Extranodal marginal-zone lymphoma (EMZL)** to predict for **Splenic marginal-zone lymphoma (SMZL)**	disease course and later malignancy	strong moderate weak none don't know	

Pairs we have missed (learn from which disease, to predict which outcomes in which disease, and how much you

would expect it to help):

Other comments:

The 24 diseases

Large B-cell lymphomas: Diffuse large B-cell lymphoma (DLBCL); Richter transformation (RT); Transformed follicular lymphoma (tFL)

Burkitt and lymphoblastic lymphomas: Burkitt lymphoma (BL); Lymphoblastic lymphoma (LBL)

Mantle-cell lymphoma: Mantle-cell lymphoma (MCL)

Indolent B-cell lymphomas: Follicular lymphoma (FL); Lymphoplasmacytic lymphoma (LPL); Extranodal marginal-zone lymphoma (EMZL); Splenic marginal-zone lymphoma (SMZL); Nodal marginal-zone lymphoma (NMZL); B-cell lymphoma, unspecified (BCL)

CLL and SLL: Chronic lymphocytic leukaemia (CLL); Small lymphocytic lymphoma (SLL)

Hairy-cell leukaemia: Hairy-cell leukaemia (HCL)

Hodgkin lymphoma: Hodgkin lymphoma (HL)

T-cell lymphomas: Peripheral T-cell lymphoma, NOS (PTCL); Angioimmunoblastic T-cell lymphoma (AITL); Anaplastic large-cell lymphoma (ALCL); T-cell lymphoma, unspecified (TCL)

Plasma-cell neoplasms: Multiple myeloma (MM); Plasma-cell leukaemia (PCL); Solitary plasmacytoma

Amyloidosis: AL amyloidosis

The 69 outcomes, by clinical domain

Each outcome is predicted at 30, 60 and 90 days and at 6, 12 and 24 months after the start of first-line treatment. Laboratory grades are based on the CTCAE laboratory thresholds; pairs can name a whole domain or "all outcomes".

Disease course and later malignancy: overall survival; second-line treatment; treatment failure; treatment failure or transformation; new haematological malignancy; new malignancy (any)

Infection and antimicrobial use: sepsis; serious infection (including invasive fungal infection and pneumocystosis); antibiotic treatment; intravenous antimicrobial use; prescription antimicrobial use; infection diagnosis or antimicrobial use

Erythroid findings and transfusion: anaemia grade 2 and 3; red-cell transfusion; any transfusion; haptoglobin decreased

Neutrophil, white-cell and platelet findings: neutropenia grade 2 and 3; leukopenia grade 2 and 3; thrombocytopenia grade 2 and 3; platelet transfusion; severe cytopenia

Lymphocyte and eosinophil counts: lymphocytosis grade 2 and 3; eosinophilia

Renal injury and support: acute kidney injury; creatinine increased grade 2 and 3; creatinine rise above 1.5 times the pre-treatment value; dialysis

Liver injury and coagulation: ALT increased grade 2 and 3; alkaline phosphatase elevation, two levels; bilirubin increased grade 2 and 3; hepatotoxicity; INR increased grade 2 and 3

Sodium and potassium: hyponatraemia grade 2; sodium below 125 mmol/L; potassium below 3.0 mmol/L; hypernatraemia grade 2 and 3; hyperkalaemia grade 2 and 3

Calcium and magnesium: hypercalcaemia grade 2 and 3; hypocalcaemia grade 2 and 3; hypomagnesaemia grade 2

Metabolic and cell-turnover markers: hypertriglyceridaemia grade 2 and 3; hypoalbuminaemia grade 2 and 3; hyperphosphataemia; hyperuricaemia; LDH increased

Cardiovascular and thrombotic events: atrial fibrillation; heart failure; stroke or transient ischaemic attack; venous thromboembolism

Acute care and other events: acute hospitalisation; emergency department stay; respiratory support; bone fracture